Name:

Class: Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.3 Particle Model of Matter

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

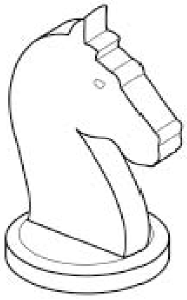
Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

01.1 A student wanted to determine the density of the irregular shaped object shown in the image below.



Plan an experiment that would allow the student to determine the density of the object.

[6 marks]

01.2 The student is given a piece of a plastic material.

The student determined the density of the material three times.

Table 1 shows the results.

Table 1

	Density in kg/m^3
1	960
2	1120
3	1040

Determine the uncertainty in the student's results.

Uncertainty = ______ kg/m^3

[2 marks]

02.1 A scientist cooled the air inside a container.

The temperature of the air changed from 20 °C to 0 °C.

The volume of the container of air stayed the same.

Explain how the motion of the air molecules caused the pressure in the container to change as the temperature decreased.

[3 marks]

02.2 The air contained water that froze at 0 °C.

The change in internal energy of the water as it froze was 0.70 kJ.

The specific latent heat of fusion of water is 330 kJ/kg.

Calculate the mass of ice produced.

Use the Physics Equation Sheet.

Mass of ice = ______ kg

[3 marks]

03 Ice cream is made by cooling a mixture of liquid ingredients until they freeze.

How do the kinetic energy and the potential energy of the particles change as a liquid is cooled and frozen?

Tick **one** box.

- ☐ Kinetic energy decreases, potential energy decreases
- ☐ Kinetic energy decreases, potential energy does not change
- ☐ Kinetic energy does not change, potential energy decreases
- ☐ Kinetic energy does not change, potential energy does not change

[1 mark]

04 **Figure 1** shows a wind turbine.

Figure 1

At a particular wind speed, a volume of 2.3×10^4 m^3 of air passes the blades each second.

The density of air is 1.2 kg/m^3.

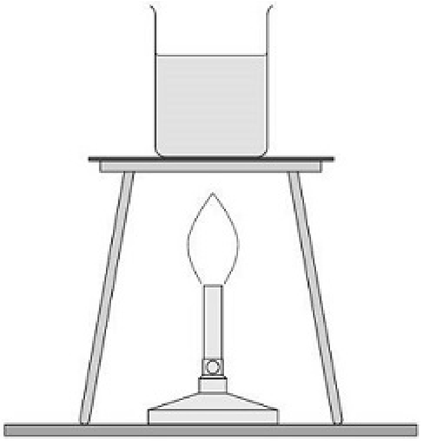
Calculate the mass of air passing the blades per second.

Mass of air per second = ____________ kg

[3 marks]

05 **Figure 2** shows a Bunsen burner heating some water in a beaker.
Eventually the water changes into steam.

Figure 2



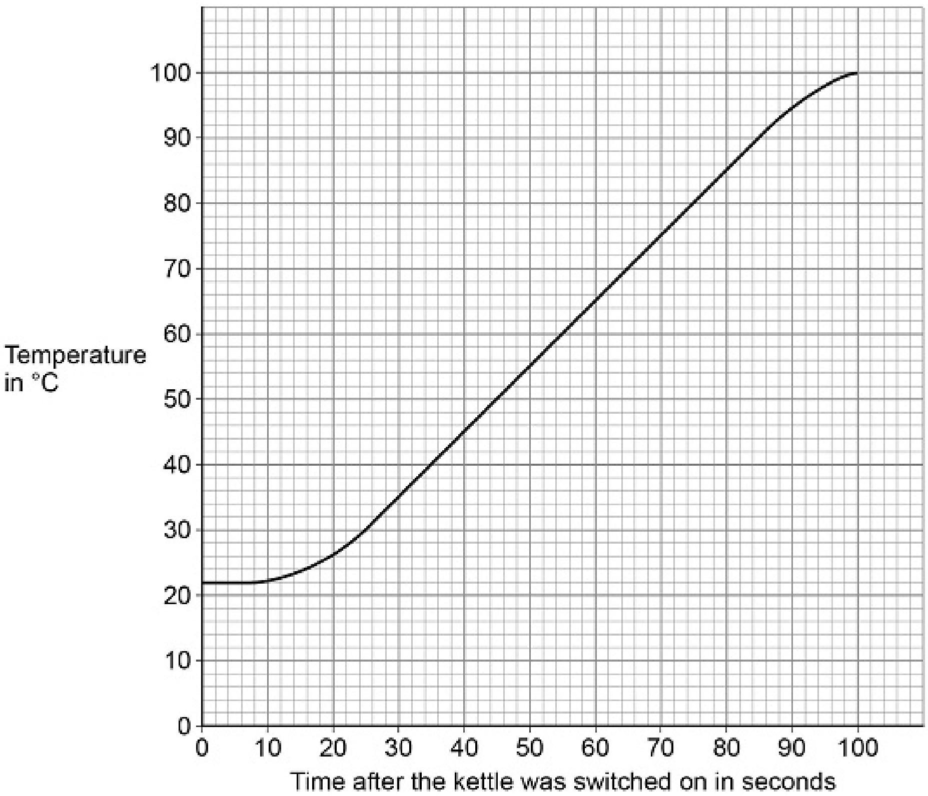
Explain how the internal energy of the water changes as it is heated from 20 °C to 25 °C

[2 marks]

06 An electric kettle was switched on.

Figure 3 shows how the temperature of the water inside the kettle changed.

Figure 3



(a) The energy transferred to the water in 100 seconds was 155 000 J.

Specific heat capacity of water = 4200 J/kg °C.

Determine the mass of water in the kettle.

Use the graph above.

Give your answer to 2 significant figures.

Mass of water (2 significant figures) = ____________ kg

(5)

(b) The straight section of the line in the graph above can be used to calculate the useful power output of the kettle.

Explain how.

(3)

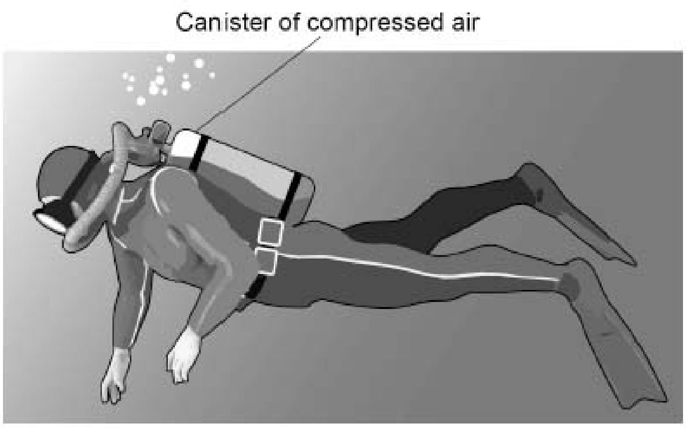
[8 marks]

07.1

Figure 4 below shows a diver.

The diver is using a canister of compressed air so that he can breathe underwater.

Figure 4

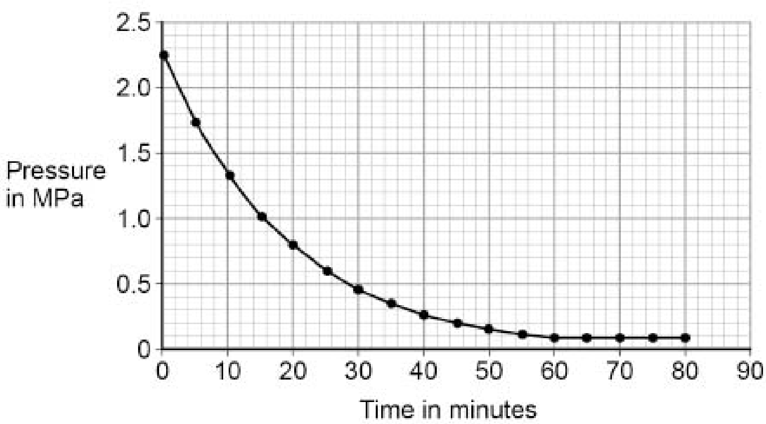


A canister of air was tested to find out how the pressure changed when it was used by a diver.

- Air was allowed to escape from the canister.
- The pressure of the air in the canister was recorded every 5 minutes for 80 minutes.

Figure 5 shows the results.

Figure 5



Estimate the atmospheric pressure.

Use **Figure 5**.

Atmospheric pressure = ____________ MPa

[1 mark]

07.2 Divers can safely stay underwater until the pressure of the air in the canister has reduced to 25% of its original value.

Determine the maximum time the diver can safely stay underwater.

Use **Figure 5**.

Time = ____________ minutes

[3 marks]

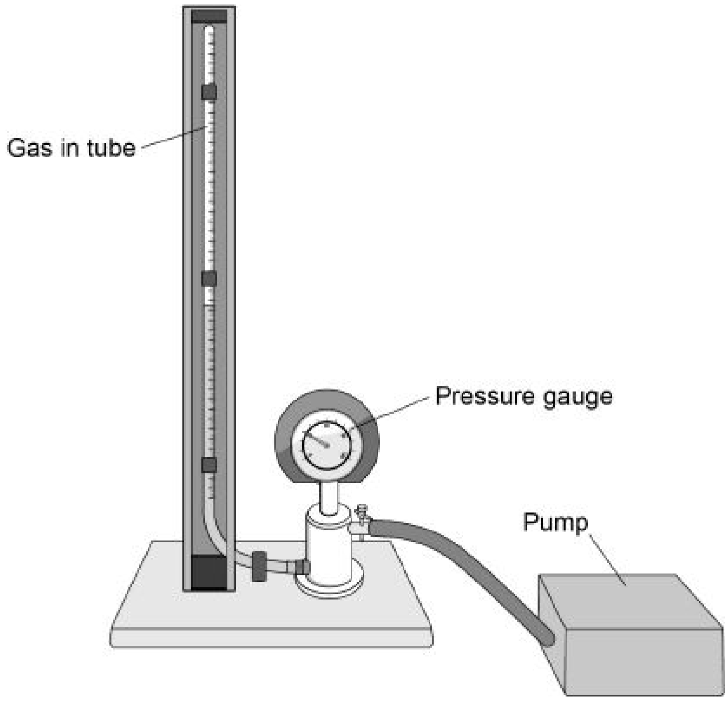
07.3 What happens to the volume of the air when it is released from the canister?

[1 mark]

08.1 A student investigated how the pressure exerted by a gas varied with the volume of the gas.

Figure 6 shows the equipment the student used.

Figure 6



A pump was used to compress the gas in a tube. As the volume of the gas decreases, the pressure of the gas increases.

If the gas is compressed too quickly the temperature of the gas increases.

Explain how the temperature increase would affect the pressure exerted by the gas.

[2 marks]

08.2 One of the student's results is given below.

pressure = 1.6×10^5 Pa

volume = 9.0 cm^3

Calculate the volume of the gas when the pressure was 1.8×10^5 Pa.

The temperature of the gas was constant.

Volume = ____________ cm^3

[3 marks]

08.3 **Figure 7** shows a person using a bicycle pump to inflate a tyre.

Figure 7

The internal energy of the air increases as the tyre is inflated.

Explain why.

[2 marks]